

Hrishikesh Kurapati

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SUMMARY

Full-Stack and AI Engineer specializing in developing robust, scalable applications and intelligent systems. Proven expertise in architecting decoupled full-stack platforms, engineering applied AI solutions with LLMs, and conducting deep learning research for bioinformatics. Adept at bridging production-grade software engineering with advanced machine learning methodologies to solve complex real-world challenges. Passionate about delivering high-performance, user-centric products driven by measurable engineering impact.

EDUCATION

Indian Institute of Information Technology , Sri City, Andhra Pradesh	2023 - Present
B.Tech, Computer Science and Engineering	CGPA: 8.73
Resonance Junior College , Telangana	2021 - 2023
Intermediate, 10-12th, Science	CGPA: 9.63

PROJECTS

FUSE-CDR — Dual-branch heterogeneous GNN framework for precision oncology May 2025 - May 2026

- Engineered FUSE-CDR, a dual-branch heterogeneous graph neural network (GraphSAGE and HGT) in PyTorch to predict cancer drug responses via biologically-guided multi-omics fusion.
- Architected an end-to-end data pipeline processing high-dimensional modalities (GDSC and CCLE datasets) using scikit-learn, RDKit, and DeepChem for topological molecular graph featurization.
- Implemented Supervised Contrastive Learning alongside BCE to enhance representation robustness, explicitly mapping matching drug-cell response pairs in the latent space.
- Achieved a peak 0.9359 AUC on the GDSC random split, outperforming foundational baselines like GraphDRP (+1.53%) and DeepTTC (+6.26%).
- Automated a rigorous 5-fold cross-validation benchmarking suite with successive halving hyperparameter optimization to validate model performance across strictly unseen inductive splits.

LeanFeastAI — Decoupled full-stack culinary platform with RAG-based generation Aug 2025 - Dec 2025

- Architected a decoupled full-stack culinary platform utilizing React 19/TypeScript for the frontend and an asynchronous Python/FastAPI backend deployed on Vercel and Render.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline leveraging LangChain, Pinecone, and the Gemini 2.5 Flash API to synthesize highly accurate, personalized recipes grounded in dietary constraints.
- Developed a stateful hands-free cooking interface integrating the Web Speech API, Picovoice intent detection, and dynamic Text-to-Speech (TTS) for voice-driven navigation.
- Designed a normalized PostgreSQL database on Supabase with Row Level Security (RLS) and real-time WebSocket synchronization for live community interactions.
- Integrated programmatic nutritional analytics and latent diffusion image generation pipelines via Spoonacular and Pollinations.ai APIs to optimize the automated meal planning experience.

CollabSync — Scalable multi-role influencer marketing and e-commerce platform Jan 2025 - May 2026

- Architected a scalable multi-role influencer marketing platform, transitioning a monolithic backend into a layered service architecture utilizing Node.js, Express.js, and MongoDB.
- Optimized database performance via advanced indexing, document embedding, and field projection, significantly reducing over-fetching across 170+ REST API endpoints.
- Reduced Brand Dashboard response latency by 95.3% (650.5ms to 30.4ms) and Influencer Explore load times by 53.8% through strategic Redis caching implementations.
- Engineered a secure authentication system with cross-site session cookies and orchestrated an end-to-end campaign lifecycle state machine incorporating real-time deliverable tracking.
- Authored a comprehensive GitHub Actions CI/CD pipeline provisioning ephemeral Redis/MongoDB containers, enforcing ESLint, and gating pull requests behind 154 automated Jest tests.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL
Frontend & Backend	React.js, FastAPI, Node.js, Express.js, Redux Toolkit, Tailwind CSS, REST APIs, WebSockets
AI, ML & Data Science	PyTorch, Hugging Face, scikit-learn, LangChain, LLMs (RAG, Fine-tuning), GNNs (PyG)
Databases & Infrastructure	PostgreSQL, MongoDB, Redis, Supabase, Pinecone, Docker, GitHub Actions, Vercel